Capstone Project Guide

Build a Real-World AI System for Your Portfolio

> ***TL;DR**: Build an intelligent GenAI system. Choose one of five tracks (or propose your own). Working interface, full documentation, honest evaluation. 40% of final grade.*

🎯 Project Overview

Industry context: Generative AI can automate creative and analytical work — but only when it is pointed at the right problem and built with proper evaluation. Most GenAI demos look impressive and fall over on the second question. This project is about building something that does not.

Key deliverables:

- Working AI system with a usable interface
- Demo video and presentation
- Complete documentation and code
- **Evaluation evidence** — how well does it actually work, and where does it fail?
- Strategic recommendations for broader market opportunities

Project scope: These systems are large. Teams should **go deep on one domain** rather than shallow across many (e.g. paper drafting for one venue only; deadline extraction from one mail platform done really well). Depth is valued over breadth.

🏞️ Project Tracks

📖 **Full descriptions:** Capstone_Project_Tracks_v4.pdf

Track	Title	In one line
1	The General Manager's Co-Pilot	An agent the factory boss talks to all day: briefings, questions, alerts, feasibility checks, actions
2	Quick-Response Subcontracting Desk	Decide which workshop gets each batch — capability, capacity, eligibility — proven against baselines in a shared simulator
3	Research Paper Ideation and Writing	Draft grounded papers **that do not read like an LLM wrote them**
4	Student Productivity Agent	Outlook / Teams agent: triage mail, extract deadlines, plan your week (**no Canvas/LMS integration**)
5	Propose Your Own	Approval required by Week 3

Choosing between them

- **Tracks 1–2** are set in **quick-response apparel manufacturing** — small batches, 7–14 day reorders, and a network of outside workshops. A small, clean dataset is provided; you can read all of it in five minutes, and **the data is deliberately not the challenge** — the agent, its tools and its interface are. **Read** `Factory_Primer.pdf` **first** — three minutes, no manufacturing background required. Everything you receive (data, dictionary, simulator, chat inbox) is listed file by file in the tracks document.

- **Tracks 3–4** are open-ended build tracks. You define the problem sharply, and you design your own evaluation. **Track 4 excludes Canvas/LMS integration** — NUS IT has confirmed that Canvas API access requires administrator approval and sits outside their normal process, so no team may use it.
- Track 2 treats the objective as a **configuration**: each team picks a primary objective of its own choosing and declares it at **Sprint 1**, but also reports at least one other objective and one hybrid (weighted) setting, so results are comparable across teams at the end of the semester.

🗓️ Timeline & Milestones

Phase	Due	Deliverables
Team formation & track selection	Week 3 — Sun, 30 Aug 2026, 11:59 PM	Team composition; track selection; Track 5 proposal. Track 2 teams choose their own **primary objective** and declare it at **Sprint 1** — not now. After the deadline, remaining students and tracks are **auto-assigned**
Sprint 1: presentation	Week 9 — Fri, 16 Oct 2026 **upload by**: 3:00 PM **presentation**: 7:00 PM	`Presentation_GroupNN_Sprint1` slides; 5-min live presentation + 2-min Q&A per group; Track 2: declare your **primary objective**
Sprint 2: presentation	Week 11 — Fri, 30 Oct 2026 **upload by**: 3:00 PM **presentation**: 7:00 PM	`Presentation_GroupNN_Sprint2` video + slides; 5-min video + 2-min Q&A per group
Sprint 3: presentation	Week 13 — Fri, 13 Nov 2026 **upload by**: 3:00 PM **presentation**: 7:00 PM	`Presentation_GroupNN` video + slides; 5-min video + 2-min Q&A per group
Sprint 3: final submission	Week 13 — Sun, 15 Nov 2026, 11:59 PM	All deliverables (`GroupNN.zip`)

> 👄 ***Filename rule — `NN` is always two digits.*** *Everywhere `GroupNN` appears below, replace `NN` with your group number **zero-padded to two digits**: Group 1 is `Group01`, Group 7 is `Group07`, Group 12 is `Group12`. So the correct names are `Presentation_Group01_Sprint1.pptx` and `Group01.zip` — **not** `Presentation_Group1_Sprint1.pptx`, `Group1.zip`, `group01.zip` or `Group_01.zip`. Capitalisation must match exactly. Non-conforming filenames lose marks.*

🏃 Sprint Reviews

The project runs as three sprints, each ending with an in-class review: **5 minutes of presentation followed by 2 minutes of live Q&A**. Sprint 1 is presented **live**; Sprints 2 and 3 use a **pre-recorded video (5 minutes)** played in

class. Timings are **hard stops — you will be cut off**. Upload your materials to your group's Canvas folder (no subfolders) **by 3:00 PM on the day of your session**, so we can run them from the lecture-theatre PC. Presentation order and session logistics are announced before each sprint.

Attendance: every team member must attend their session in **Sprints 1, 2 and 3** and be ready to **answer in the Q&A** — but not everyone needs to present or speak in the video. Absence without a valid reason (MC by email) affects your individual mark.

Sprint 1 review — Week 9 (Fri, 16 Oct 2026)

Objective: show that your plan survives contact with reality — demonstrate early progress, surface your assumptions, and gather feedback while there is still time to change course. This round is a warm-up for presentation experience and is evaluated lightly.

Format: a **live 5-minute presentation** (no video for this sprint) + 2-minute Q&A, covering, clearly and in order:

1. **Problem statement & proposed solution** — including an architecture diagram (the example deck shows an agentic-LLM architecture)
2. **Business value**
3. **Sprint 1 progress** *(optional)* — early wins, if you have them
4. **Risk assessment**
5. **Key assumptions** (if any)
6. **Sprint 2 development roadmap** — including your success criteria

Track 2 teams: this is where you **declare your primary objective** — minimise lateness, minimise defects, keep allocation fair across workshops, or stay robust when a workshop fails — and say why you chose it. The objective is **your team's own choice, not something we assign**; it belongs with your problem statement and success criteria above.

Materials required: the slides, named `Presentation_GroupNN_Sprint1` (`.pptx` or `.pdf`, `NN` = two-digit group number, e.g. `Presentation_Group01_Sprint1.pptx`, `Presentation_Group12_Sprint1.pdf`), uploaded by **3:00 PM, Fri 16 Oct 2026**. See `DSS5105_Sprint_Review_Example.pdf` (Week 0 → Capstone Project) for an example of slide structure and flow.

Sprint 2 review — Week 11 (Fri, 30 Oct 2026)

Objective: show a working system taking shape and a credible plan to prove it works. From Sprint 2 onward, evaluation is **more rigorous** — technical depth, validation, and how clearly you present.

Format: a **pre-recorded video (5 minutes)** played in class + 2-minute Q&A. The video must cover:

1. **Objective** (~1 min) — the problem and who it serves
2. **Tech section** (~2 min) — architecture overview; implemented and planned features, each with an owner and a status (done / in progress / planned); and your **validation plan** — how you will show the system works (e.g. ≥*30-question evaluation set verified against the data files*)
3. **Demo** (~2 min)

Materials required: the video plus the slides behind it, both named `Presentation_GroupNN_Sprint2` (`NN` = two-digit group number; the extension tells them apart: `.mp4` vs `.pptx` / `.pdf` — e.g. `Presentation_Group01_Sprint2.mp4` and `Presentation_Group01_Sprint2.pptx`), uploaded by **3:00 PM, Fri 30 Oct 2026**. **First slide must show**: group number, team members with student IDs, public GitHub link, demo link — and check the links actually open.

Sprint 3 review — Week 13 (Fri, 13 Nov 2026)

Objective: the finish line — final presentation and complete handover of everything you built, with honest evidence of how well it works.

Format: same as Sprint 2 — a **pre-recorded video (5 minutes)** played in class, followed by **2 minutes of live Q&A**. Suggested flow within the video:

1. **Objective** (~1 min) — the problem, the users, why it matters
2. **Demo** (~2 min)
3. **Capabilities, evidence and limitations** (~1.5 min) — you must cover all three:
   - **Core capabilities** — what your system does well
   - **Evidence** — *how you know.* Numbers from your evaluation, not adjectives
   - **Limitations** — what breaks it, which inputs produce poor results, what you would fix
4. **Conclusion** (~0.5 min) — achievements, lessons learned

> *Teams that present only successes score poorly. A team that says "our agent answers aggregation questions at 91% accuracy but fails on comparative questions — here is the failure analysis" demonstrates far more understanding than one claiming everything works.*

Materials required: the video, named `Presentation_GroupNN.mp4`, plus the slides behind it (`Presentation_GroupNN.pptx` / `.pdf`) — `NN` = two-digit group number, e.g. `Presentation_Group01.mp4` and `Presentation_Group01.pptx`. Uploaded by **3:00 PM, Fri 13 Nov 2026**. **First slide must show**: group number, team members with student IDs, public GitHub link, demo link. Non-conforming filenames (including the two-digit padding and capitalisation) lose marks. The final submission `GroupNN.zip` follows by **Week 13, Sun 15 Nov 2026, 11:59 PM** — see **Final Deliverables** below.

🎯 Final Deliverables

The final presentation itself is described under **Sprint 3** above. What remains is the final submission.

Final submission (Week 13, Sun 15 Nov 2026, 11:59 PM)

One zip named exactly `GroupNN.zip`, where `NN` is your group number zero-padded to **two digits** — `Group01.zip`, `Group07.zip`, `Group12.zip` (not `Group1.zip`). It contains:

File	Description
`Readme.pdf`	Setup instructions, requirements, how to run it locally
`Slide.pdf` / `Slide.pptx`	Final deck — the same deck presented at Sprint 3
`Video.mp4`	Demo video (not the Sprint 3 presentation video)
`Documentation.pdf`	Detailed documentation, max 15 pages
`Evaluation.pdf`	**Evaluation report — see below**
`Teams.pdf`	Roles and contribution breakdown
`Strategic_Recommendations.pdf`	Strategic analysis

📄 **`Documentation.pdf` (max 15 pages)**

1. **Team information** — group number, members and contributions, public GitHub link, demo link
2. **Problem statement** — the problem, target users, why existing solutions fall short

3. **System architecture** — design, components, data flow
4. **Technical stack** — technologies used and *why those choices*
5. **Features and capabilities** — with implementation insights
6. **Challenges and limitations** — failure modes, edge cases, mitigations
7. **References** — papers, posts and repos that informed you, and *how each contributed*
8. **Conclusion** — achievements, lessons, future work

`Evaluation.pdf` — new this year, and it carries real weight

Every track must answer: *how do you know your system works?*

Track	Minimum evaluation requirement
1	A **tool specification** (name, input, output, purpose, non-goals, failure mode for every tool) **and** a **≥30 question evaluation set** with answers verified against the data files, including ≥5 ambiguous / unanswerable / hallucination-bait questions and ≥5 feasibility or action requests. State how you verified the answers
2	Run the shared simulator (`track2/harness/simulate.py`); report your primary objective, at least one other objective and one hybrid (weighted) setting, **against all three baseline allocators**, including at least one `--shock` run; **report the cases where a baseline wins** and what your system adds over a simple rule-based allocator (no LLM, about the size of the shipped baselines). **Plus**: results on all 30 requests in `track2/data/dispatch_requests.txt`, by behaviour (extract / clarify / refuse / decline-to-answer), judging the decision and the faithfulness of the explanation separately
3	*Example*: AI-writing detector scores before/after revision; **human study with ≥3 participants**, accuracy reported honestly
4	*Example*: **time saved** — how many hours per week, and on what (inbox triage, deadline tracking, planning) — measured against doing the same tasks manually

Tracks 1 and 2 requirements are fixed. For **Tracks 3 and 4** the rows above are **examples** — you may propose your own evaluation of equivalent rigour instead (state it in your Sprint 2 validation plan).

Required content:

1. **What you measured and why that metric**
2. **How the evaluation set was built** — and why it is not rigged in your favour
3. **Results**, including per-category breakdown
4. **≥10 failure cases analysed individually** — what went wrong and why
5. **What you would measure with more time**

> ***Do not tune on your evaluation set and then report the score.*** *If you used it for development, say so and report a held-out result as well. Honesty here is graded; we can tell, and inflated numbers cost more marks than low ones.*

`Teams.pdf`

Individual responsibilities, contribution percentages, specific tasks completed, collaboration dynamics.

`Strategic_Recommendations.pdf`

> "*What other directions could GenAI help industries? Suggest recommendations and identify 3 low-hanging fruit opportunities (few people doing it, highest impact, lowest implementation effort).*"

- Focus on specific industries and use cases
- Back recommendations with real market research

- Concrete implementation approaches
- **Generic ChatGPT-generated responses score near zero.** Higher marks for unique insights backed by real evidence

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🏆 Grading Criteria

Category	Weight	Focus
Problem solving	35%	How well the core capabilities are actually addressed
Evaluation quality	15%	Rigour and honesty of your evidence; depth of failure analysis
User interface design	15%	UX-first; usability matters more than aesthetics
Engineering quality	20%	Clean code, reliability, scalability
Documentation	8%	Clarity of README, comments, system docs
Presentation	7%	Demo quality, storytelling, Q&A

🤝 Peer reviews

Optional peer review near submission. Individual marks may be adjusted so that contribution is fairly reflected.

🎯 Success Tips

Do:

- ✅ Build the simplest version that demonstrates the core capability, then improve it
- ✅ **Write your evaluation set before you write the system** — it defines what "working" means
- ✅ Document as you build
- ✅ Use version control with meaningful commits

Don't:

- ❌ Over-engineer before the basics work
- ❌ Ignore error handling — demos fail live
- ❌ Chase non-core features (more languages, prettier themes) while the core is weak
- ❌ Report only your successes
- ❌ Let an LLM do arithmetic that a query should do (Tracks 1 & 2: this is the #1 source of wrong answers — that is what tools are for)

✅ Submission Checklist

Sprint 1 (Week 9, Fri 16 Oct 2026, 3:00 PM)

- ☐ `Presentation_GroupNN_Sprint1` slides uploaded — two-digit group number, e.g. `Presentation_Group01_Sprint1.pptx`
- ☐ Live presentation rehearsed and within the 5-minute limit
- ☐ Every member attending and ready for Q&A

Sprint 2 (Week 11, Fri 30 Oct 2026, 3:00 PM)

- [ ] `Presentation_GroupNN_Sprint2` video and slides uploaded — two-digit group number, e.g. `Presentation_Group01_Sprint2.mp4`
- [ ] Video tested end to end and within the 5-minute limit
- [ ] GitHub and demo links on the first slide verified public and working
- [ ] Every member attending and ready for Q&A

Sprint 3 presentation (Week 13, Fri 13 Nov 2026, 3:00 PM)

- [ ] `Presentation_GroupNN.pptx/.pdf` and `Presentation_GroupNN.mp4` uploaded — two-digit group number, e.g. `Presentation_Group01.pptx`
- [ ] Video tested end to end and within the 5-minute limit
- [ ] GitHub and demo links verified public and working
- [ ] Every member attending and ready for Q&A

Final submission (Week 13, Sun 15 Nov 2026, 11:59 PM)

- [ ] `GroupNN.zip` with all seven files, exact filenames — two-digit group number, e.g. `Group01.zip`, `Group12.zip`
- [ ] `Evaluation.pdf` includes ≥10 analysed failure cases
- [ ] Repository public with clean history
- [ ] System runs from a clean checkout following your README

🚀 Why This Matters

Every one of these tracks is a real problem someone is paying to solve right now. Tracks 1 and 2 are live problems in an actual factory — the manager really does arrive at 7 a.m. without answers, and the allocation really is decided in a group chat. Build something that would survive contact with them.

📚 Additional Resources

- Tracks 1–2: your track folder (`track1/` or `track2/`) — primer, data, dictionary, simulator · agent frameworks (LangGraph, AutoGen, CrewAI)
- Track 3: AI-Scientist · Wikipedia *Signs of AI writing*
- Track 4: Microsoft Graph API docs · Google Workspace API docs